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Effect of Oxygen on Colloidal Stability of Titanium Nitride Nanoparticles Synthesized by Laser Ablation in Liquids

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Abstract—The effect of oxygen existing in an ablation medium during synthesis of titanium nitride (TiN) nanoparticles (NPs) by pulsed laser ablation in liquid (PLAL) on colloidal stability of obtained solutions was studied. It was shown that an increase in the oxygen content both incorporated in liquid molecules and in the form of dissolved gas increases the colloidal stability of synthesized NPs. The results obtained extend the range of available methods for developing new nanomaterials due to control of colloidal stability of laser-synthesized NPs.

Keywords: laser ablation in liquid, colloidal stability, nanoparticles, titanium nitride

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1. INTRODUCTION

Pulsed laser ablation in liquid (PLAL) is a powerful actively developing tool for synthesizing ultrapure colloidal solutions of various nanoparticles (NPs) [1–3]. Nanomaterials synthesized through its use gained wide acceptance in biomedicine [4, 5], catalysis [6], and other fields [7]. The key advantages of the method are its implementation simplicity, purity of produced nanomaterials [8], and the feasibility of synthesizing NPs with unique shapes and compositions [9]. One of the promising nanomaterials produced by the PLAL method are titanium nitride (TiN) NPs [10]. These NPs feature a plasmon peak in the transparency window of biological tissues (between 650 and 800 nm), which allows their application to photon therapy [11] and bioimaging based on photoacoustics [12, 13].

One of the most important parameters controlling the NP applicability to biomedicine is their colloidal stability affecting both NP toxicity and biodistribution in an organism [14]. The effect of synthesis parameters on colloidal stability of TiN NPs produced by PLAL have not yet been adequately studied.

In this work, we study for the first time the effect of oxygen existing in the ablation medium during TiN NP synthesis by the PLAL method on the colloidal stability of produced solutions.

2. METHODS AND MATERIALS

The schematic diagram of the experimental setup is shown in Fig. 1a. A TiN target (GoodFellow, 99.5% purity) was exposed to fs laser radiation (1030 nm, 270 fs, 30 μJ, 100 kHz) focused by a 100-mm F-Theta objective. The liquid layer along the optical path was 3 mm, the liquid volume was 10 ml, and the duration of one synthesis was 10 min. During PLAL, the laser beam scanned the target surface with a speed of 4 m/s.

To remove dissolved gaseous oxygen from liquid, the latter was "pumped" by gaseous nitrogen N_2 as shown in Fig. 1b. To this end, a tube was immersed into a cell with target, through which nitrogen was fed with a rate of 5 L/h. Pumping was began 5 min before the ablation start and continued during the entire synthesis.

Size and morphological properties, as well as the chemical composition of synthesized NPs were measured using scanning electron microscopy (SEM) and energy-dispersive X-ray analysis (EDX) using

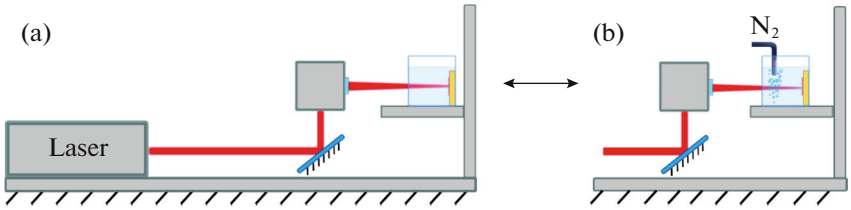

Fig. 1. (a) Schematic of the experimental setup for PLAL. (b) Experimental scheme with liquid pumping by nitrogen N_2.

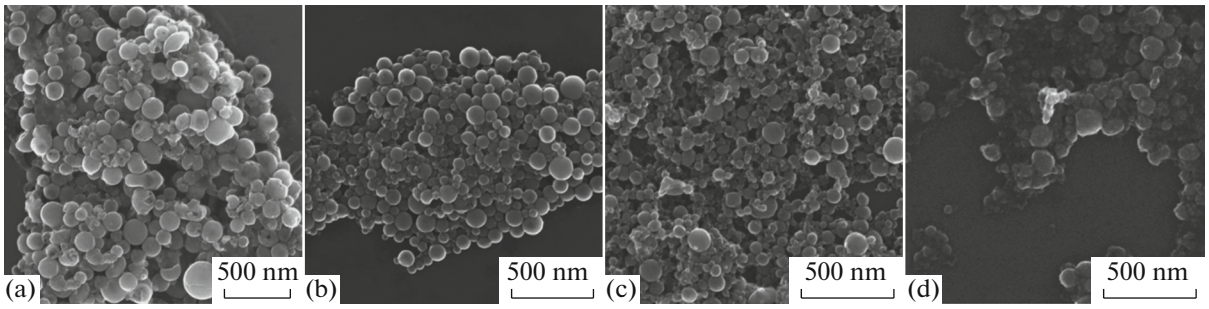

Fig. 2. SEM images of TiN NPs synthesized by the PLAL method in (a) hydrogen peroxide, (b) water, (c) acetone, and (d) hexane.

a MAIA 3 (Tescan) microscope with an X-Act EDX detector (Oxford Instruments), operating at an accelerating voltage of 20 kV. Optical extinction spectra were measured using an ML 122 spectrophotometer (SOL Instruments) with cells of 10-mm optical path. The concentration of produced colloidal solutions was measured by the atomic absorption spectroscopy method (the titanium content in samples was detected) using a Lumex MGA-1000 device.

3. RESULTS AND DISCUSSION

To study the effect of oxygen on the colloidal stability of NPs produced by the PLAL method, a number of liquids with various oxygen atom contents in molecules of these liquids were used as media for synthesis: hydrogen peroxide (H_2O_2), water (H_2O), acetone (C_3H_6O), and hexane (C_6H_{14}). PLAL led to rapid formation of colloidal solutions. Figure 2 shows the SEM images of NPs synthesized in each of liquid.

As seen in these images, NPs were morphologically homogeneous and spherical. A statistical analysis of sizes of produced NPs showed that they in all cases featured a log-normal distribution with average sizes from 44 to 98 nm. Individual parameters of NP size distributions and concentrations are listed in Table 1.

The colloidal stability was determined using the method for measuring the kinetics of the optical extinction spectra of produced NPs. Figure 3a shows the typical picture of time variations in optical spectra for NPs synthesized in acetone. The optical extinction uniformly decreased in the whole spectral range, and the spectral shape and plasmon peak position were unchanged. As a result, the extinction of TiN NP solutions was directly proportional to the NP concentration. We note that similar kinetics (uniform decrease) of optical spectra was observed in other liquids. Figure 3b shows the optical extinction

Table 1. Properties of NP produced in various liquids

Medium	Average NP size, nm	Root-mean-square deviation NP size, nm	Concentration, μg/ml
H_2O_2	98	41	82
Water	58	25	67
Acetone	44	17	59
Hexane	45	19	63

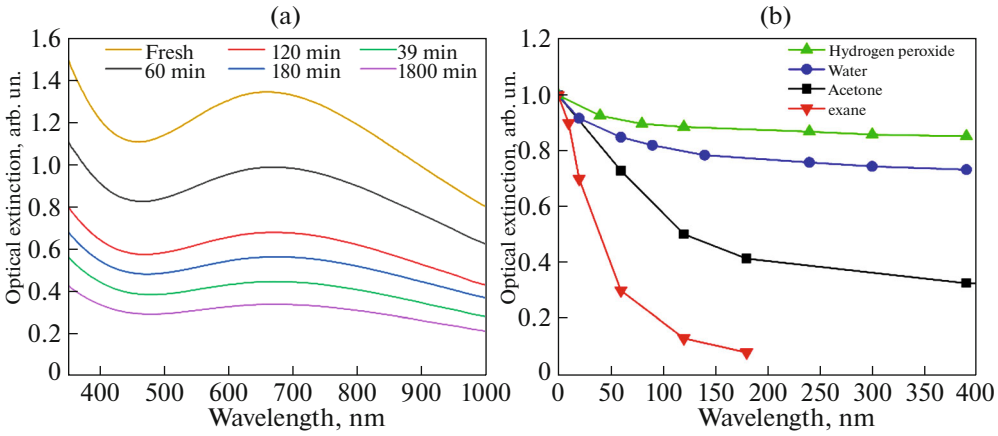

Fig. 3. (a) Variations in the spectra of optical extinction of TiN NPs synthesized in acetone. (b) Normalized kineticses of optical extinction at the plasmon resonance wavelength of TiN NPs synthesized in liquids with various oxygen contents.

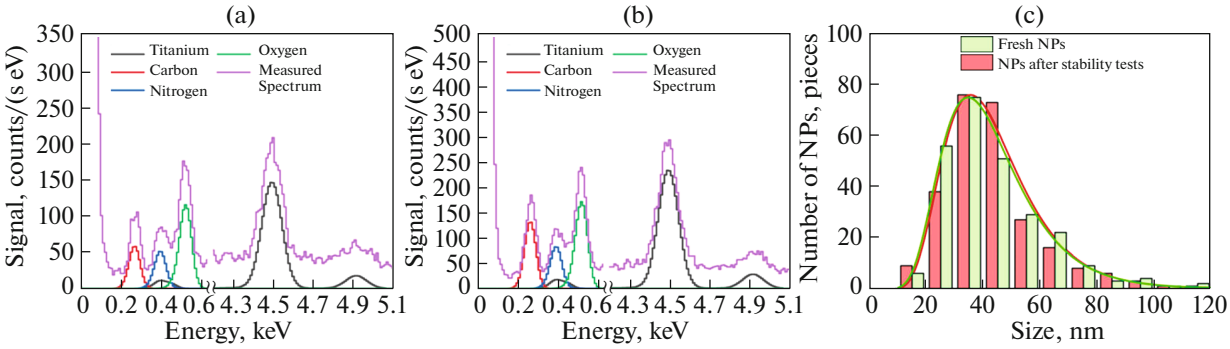

Fig. 4. (a) EDX spectrum of TiN NPs immediately after synthesis in acetone and (b) after testing the colloidal stability, as well as (c) size distributions of NPs.

kinetics for all synthesized NPs. As follows from these data, optical extinction drop rate increased with decreasing oxygen atomic fraction in liquid molecules.

The changes in extinction spectra of colloidal solutions can be caused by changes in properties of NPs themselves (size, shape, and chemical composition) and changes caused by limited colloidal stability of NPs. To ascertain which of the two mechanisms was responsible for changes in the TiN NP extinction, NP properties were measured immediately after synthesis and after measurements of optical spectrum kinetics. The results for NPs synthesized in acetone are shown in Fig. 4 (the results for other NPs were similar and are not shown). Neither morphological properties (the data are not presented), nor chemical composition (Figs. 4a and 4b), nor the size distribution (Fig. 4c) exhibit noticeable changes for this time. These results eliminate the effect of change in properties of individual NPs on a decrease in the optical extinction of colloidal solutions of TiN NPs.

Thus, the decrease in the optical extinction was caused by limited colloidal stability of TiN NPs. The invariance of the spectral position of the plasmon peak (Fig. 3a) excluded NP coagulation into dense aggregates from possible causes of changes in optical spectra; thus, the curves in Fig. 3b correspond to the kinetics of colloidal solution concentrations. One more confirmation of the above conclusion is the precipitate appearance on the bottom of vessels for all liquids except for H_2O_2.

We understand the colloidal stability as the ability of NPs dispersed in liquid to remain in suspension under Brownian motion. Individual inorganic NPs can be colloidally stable in aqueous solutions at sizes less than 100—200 nm. As seen in Table 1, average sizes of all NPs produced within this study were smaller than 100 nm; therefore, all of them should be colloidally stable, which is not consistent with experimental results. Furthermore, the largest NPs produced by laser ablation in H_2O_2 (the average size is 98 nm)

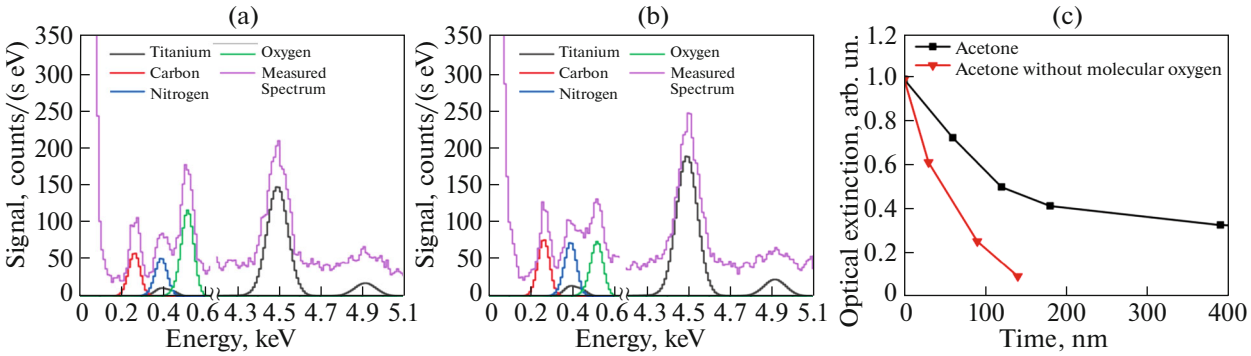

Fig. 5. (a) EDX power of NPs synthesized in acetone (b) with gaseous oxygen and without it, as well as (c) kinetics of their optical extinction at the plasmon resonance wavelength.

appeared most stable. The found inconsistency is explained by partial NP coagulation into low-density aggregates with interlayers, which resulted in their precipitation with time.

Physicochemical processes occurring during the NP formation by the PLAL method result in the formation of the same electric charges of synthesized NPs, which provides their electrostatic stabilization [15]. The parameters which affecting the electrostatic stabilization efficiency are: the electric potential near the NP surface and the Hamaker constant and properties of liquid in which NPs are [16]. We assume that an increase in the stability of NPs synthesized by laser ablation of the TiN target in liquids with increased oxygen contents is caused by partial oxidation of TiN by oxygen existing in liquid, to titanium oxides and oxynitrides. Hydroxiling of these compounds followed by proton loss and the TiO^- or $TiNO^-$ formation on the surface by the mechanism similar to that described in [15] leads to the formation of a strong negative charge on the NP surface, which provides their electrostatic stabilization. An alternative cause of the increase in the stability can be a potential decrease in the Hamaker constant for partially oxidized particles. However, it should be noted that the NP colloidal stability can also be affected by properties of liquids themselves (concentration and valence of conductivity ions and permittivity) in which synthesis was performed and the colloidal stability was measured.

To eliminate the effect of differences in properties of liquids on the results, an additional experiment was performed, in which NPs were synthesized in the same liquid (acetone), but with different contents of dissolved molecular oxygen O_2. After removal of molecular oxygen from liquid, NP synthesis was performed in it, and, then, the chemical composition of produced NPs and concentration kinetics were evaluated. Removal of dissolved molecular oxygen from liquid decreased the oxygen content in synthesized NPs (Figs. 5a and 5b) by a factor of 2.2. The optical extinction kinetics (Fig. 5c) shows a significant decrease in the NP colloidal stability when removing dissolved molecular oxygen from liquid.

Thus, even during NP synthesis in the same liquid, a decrease in the oxygen content in this liquid significantly lowers the NP colloidal stability. These data allow drawing the conclusion that an increase in the oxygen content in liquid (both in liquid molecules and as dissolved gas) increases the colloidal stability of NPs produced by laser ablation of TiN targets in this liquid. However, the specific mechanism of increasing the colloidal stability remains unknown. This mechanism is a subject of future studies.

4. CONCLUSIONS

Colloidal solutions of TiN NPs were synthesized by the PLAL method in liquids with various oxygen contents in liquid molecules themselves and in the form of dissolved gas. It was shown that an increase in the oxygen content in liquid leads to a significant increase in the colloidal stability of produced solutions.

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